David Liu

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Education

Bachelor of Science (BS), Engineering – Harvey Mudd College, Claremont, CA, USA 2028 | GPA: 3.741

Experience

Electro-Mechanical Engineering Intern May 2026 – August 2026
MRP Solutions *Plattsburgh, NY, USA*

- Performed root-cause analysis on 15+ electro-mechanical failures across automated manufacturing systems, identifying electrical, mechanical, and process issues to support corrective actions.
- Conducted testing, calibration, and validation on 5+ manufacturing systems, identifying performance deficiencies and supporting engineering modifications to improve operational reliability.
- Designed 10+ mechanical assemblies, machine guards, and access-control systems in SolidWorks for automated manufacturing equipment, producing fabrication-ready CAD models and engineering drawings.

Involvement

Electro-Mechanical Lead – Mudd Robo-Sub Team, Harvey Mudd College August 2024 – Present

- Redesigned PCB layouts to improve signal integrity, reduce electrical noise, and enhance subsystem reliability in underwater environments.
- Designed and prototyped 3+ iterations of a torpedo release mechanism for an autonomous underwater vehicle, improving deployment reliability and repeatability under submerged conditions.
- Performed FEA in SolidWorks to evaluate stress concentrations and deformation under operational loading across 5+ design configurations, guiding structural optimization and improving robustness.

Propulsion Lead – Mudd Amateur Rocketry Club, Harvey Mudd College August 2024 – Present

- Led mechanical development of Apollyon I, a full-M-class FAR Unlimited competition rocket, coordinating 40+ members across propulsion, structures, avionics, recovery, and simulation to achieve 1st Place at FAR Unlimited 2026.
- Directed systems integration and launch-readiness reviews across 5 subteams, driving design decisions throughout the development cycle.
- Evaluated 10+ configurations for motor selection, stability margins, recovery sizing, and composite airframe design using OpenRocket, RASAero II, MATLAB, SolidWorks, and COMSOL.
- Manufactured and integrated flight hardware for Gladius III, contributing to a 3rd Place finish (2025) and validating simulations against flight data with 99.95% apogee accuracy.

Projects

Autonomous Underwater Robot with Multi-Sensor Environmental Monitoring January 2026 – May 2026
Harvey Mudd College *Project Link*

- Developed analog signal-conditioning circuitry and closed-loop depth control on a Teensy 4.0, integrating MATLAB regression models with Arduino-based feedback control.
- Designed and integrated pressure, pH, turbidity, and temperature sensing systems for an autonomous underwater vehicle capable of collecting environmental data to depths of 2 m.
- Conducted field testing to validate sensor performance and analyze environmental trends across turbidity, pH, temperature, and depth measurements.

Convertible Sofa-Bed System for Camper Van Applications January 2025 – May 2025
Harvey Mudd College *Project Link*

- Led the end-to-end design of a deployable sofa-bed system for a camper van client, generating 5+ design concepts while balancing ergonomics, manufacturability, structural performance, and storage constraints.
- Evaluated 3 support-leg architectures and multiple hinge configurations through prototype testing and FEA-driven optimization to increase stiffness and reduce peak stresses.
- Created CAD models, fabricated a full-scale prototype, and validated system performance through deployment-force testing and full configuration transitions completed in 82 seconds.

Skills

Electrical & Embedded: Circuit Design, KiCad, PCB Layout, Analog Signal Conditioning, Electrical Fault Analysis, Sensor Integration, Embedded Systems Programming, Arduino, Teensy, LabVIEW

Programming: Python, MATLAB, C++, Simulink, Git

Simulation & Analysis: COMSOL, FEA, CFD, Data Analysis, Flight Simulation, OpenRocket, RASAero II

CAD & Design: SolidWorks, Engineering Drawings, GD&T, DFM, Mechanical Design, Materials Selection

Fabrication: Machining, CNC Machining, 3D Printing, Mechanical Assembly, Prototyping, Composite Layups, Laser Cutting, Waterjet Cutting

Certifications: NAR Level 1 High-Power Rocketry